Embodied Neural Rendering in Self-Aware Networks

A Falsifiable Model of Vision, Sensorimotor Feedback, Body Maps, and the Gamma Consideration Sandwich

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Abstract

Biological vision is not a passive sequence of image classifications. It is a recurrent sensorimotor process in which retinal input, internally generated expectations, eye and body movements, corollary discharge, proprioception, touch, vestibular input, and interoception jointly constrain an evolving body-and-world state. Self-Aware Networks (SAN) interprets this closed-loop construction as **Neural Rendering**. Its proposed mechanism, Neural Array Projection Oscillation Tomography (NAPOT), hypothesizes that receiver-relative phase relationships, spatial wave geometry, and recurrent partial projections contribute routing information beyond rate, power, static connectivity, scalar synchrony, and unconstrained recurrent history.

This paper develops the embodied-vision case in greater detail. It distinguishes neural arrays from dendritic input arrays; models a dendritic arbor as a branch-preserving receive-transform-re-express interface; and uses overlapping Gaussian radial-basis functions as an engineering representation of spatially patterned dendritic state without positing a literal internal screen or observer. It then formalizes the **Gamma Consideration Sandwich (GCS)** as a falsifiable hypothesis: phase-structured laminar interactions regulate when sensory evidence, recurrent expectations, and motor or efference-copy context become mutually effective in updating an independently measured body-related latent such as perceived limb position, agency, ownership, or action recalibration.

The framework is tested through body-map cross-decoding, closed-loop virtual-body delay, layer-resolved model discrimination, phase-specific perturbation and rescue, branch-geometry ablation, social-cue conflict with active sampling, and harmonic controls for proposed 20-to-40/60/80-Hz transitions. Conditional surprise and Bayesian information gain are separated from perceptual clarity, semantic relevance, and causal effectiveness. No SAN, GCS, NAPOT, or consciousness result is reported. The paper defines a discriminating biological research program and a dated 2011-2022 developmental genealogy rather than claiming validation.

maps; dendritic computation; neural oscillations; Gamma Consideration Sandwich; NAPOT; Self-Aware Networks; neural rendering

Figure status: Figures 1-7 are author-directed explanatory schematics. They are not anatomical atlases, neuroimaging results, or empirical confirmation of SAN. Observed components, SAN hypotheses, engineering analogies, and provenance are separated inside the figures; where compressed figure wording differs in specificity, the manuscript's definitions, controls, and limitations govern.

Keywords: active vision; sensorimotor integration; corollary discharge; proprioception; body

Scope and evidence boundary

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 ### Version 1.2 illustrated source-correction candidate - 2026-07-13 Theory-and-methods paper. This manuscript reports no new human or animal experiment. This illustrated candidate has not been deposited or published as Version 1.2.

1. Introduction: vision is an embodied control loop

A camera can record a sequence while remaining unchanged by what it records. A biological visual system cannot. Eyes move, pupils change, the head and body reposition, actions alter the scene, and the consequences of those actions return through several sensory channels. What is visible at the next moment depends partly on what the organism just did. A useful theory of visual learning must therefore explain not only feature recognition, but how an agent separates self-caused from externally caused change, maintains object and body state across movement, predicts consequences, and revises its model when prediction and reafference disagree.

The problem is especially clear during active perception. A stable visual estimate must survive saccades, head and limb movement, self-occlusion, changing viewpoints, and delayed sensory consequences. The organism must infer persistent entities, distinguish external change from self-caused change, and revise its body-and-world estimate when predicted and observed consequences diverge. These demands make a closed-loop account more appropriate than a camera or single-pass classifier metaphor.

Self-Aware Networks (SAN) proposes that recurrent biological computation continually constructs a distributed state that is useful to later perception and action. SAN calls this operation **Neural Rendering**. The word *rendering* does not imply a picture presented to an inner spectator. It names a causal state update in which distributed inputs are transformed and re-expressed to downstream receivers. NAPOT is the stronger, conditional proposal that receiver-relative timing and multiple partial projections support an identifiable reconstruction of a latent body-and-world state. If an identifiable inverse operation cannot be shown, *oscillatory recurrent integration* is the more accurate description.

This paper does not repeat the full SAN theory. It isolates one demanding case: embodied vision. It asks whether a defined chain from action, through prediction and multisensory reafference, to a measurable body-state update can distinguish the GCS / NAPOT proposal from ordinary recurrent processing, rate and power models, and a preparation-backed alternative ordering of feedforward and feedback frequencies.

The paper makes five contributions:

1. It separates established visual, motor, body-map, and dendritic components from SAN-specific synthesis.
2. It preserves the functional genealogy of Micah Blumberg's 2011, 2012, 2017, and 2022 work while keeping later terminology at its true date.
3. It defines an explicit visual-motor-reafferent causal sequence and independent body endpoints.
4. It turns the dendritic-screen and GCS metaphors into geometry-preserving models and failure tests.
5. It defines matched model comparisons, causal interventions, and failure conditions that can narrow or reject the proposed biological mechanisms.

Figure 1. Active recurrent visual-sensorimotor architecture. The retina-optic-nerve-LGN-V1/V2 route, posterior visual streams, motor pathways, cerebellar forward modeling, and modality-specific sensory returns summarize established anatomical constraints at schematic resolution. SAN adds the hypothesis that receiver-relative phase structure coordinates the distributed update; the figure does not posit an inner observer or identify one cortical site as the rendered scene.

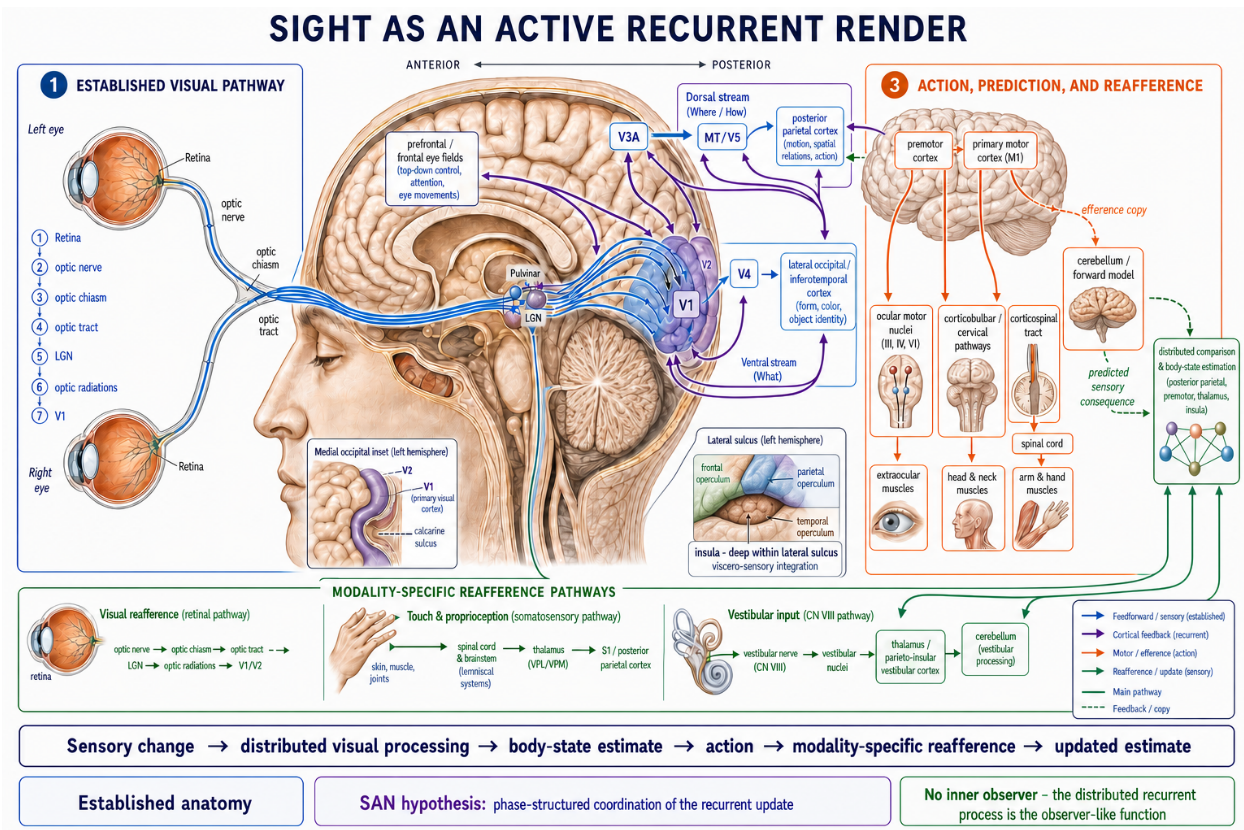

Figure 1: Sight as an active recurrent render

2. Established biological constraints and exact boundaries

2.1 Distributed and recurrent visual organization

Vision begins with structured transformation rather than raw pixel copying. Retinal circuits perform feature- and motion-sensitive operations before signals reach cortex. Thalamic and cortical pathways preserve spatial organization while transforming contrast, orientation, color, motion, depth, boundaries, and task context. The hierarchy is neither a single feedforward chain nor an undifferentiated recurrent network. Ascending, lateral, and descending pathways interact across several temporal scales. The anatomical hierarchy mapped by Felleman and Van Essen (1991) and the predictive-coding model of Rao and Ballard (1999) are important prior frameworks; neither is evidence for NAPOT by itself.

The ventral and dorsal pathways provide a useful but simplified distinction. Ventral processing emphasizes object identity and relatively transformation-tolerant recognition; dorsal and parietal processing emphasizes position, movement, coordinate transformation, attention, and visually guided action. Both streams contain mixed information and recurrent cross-talk. In this paper they motivate separable *identity / world* and *spatial / action* variables, not two isolated software modules or a claim that every cortical signal belongs to one stream.

Recurrent processing matters most when the input is ambiguous, cluttered, partial, or altered by the observer's own movement. Yet recurrence alone is not the SAN delta. A recurrent network can solve a task without phase-specific routing, multiple-projection reconstruction, a body latent, or an action-dependent error. It is therefore a required comparator rather than a result SAN may claim as its own.

2.2 Active sensing, corollary discharge, and predicted reafference

Saccades are information-gathering actions. Each changes the retinal image, so stable behavior requires the nervous system to relate expected self-caused displacement to new sensory evidence. Sommer and Wurtz (2002) identified a defined corollary-discharge pathway contributing to visual updating in a sequential-saccade preparation. That result establishes a bounded mechanism, not a universal embodiment circuit.

The broader control logic is nevertheless well motivated. A selected motor command can be copied into a forward model; the model predicts the sensory consequences of the action; movement changes retinal, proprioceptive, tactile, vestibular, and interoceptive signals as applicable; and the observed return can be compared with the prediction. A mismatch can alter the pose estimate, ownership or agency judgment, action calibration, world model, or the next policy. These endpoints must be measured separately. Calling every post-action difference a *prediction error* would hide the important question of which variable changed and which circuit used it.

Related active-inference work treats perception and action as coupled operations under generative models rather than as passive registration [32]. That literature is a relevant comparator, not evidence for SAN by itself. Recent reviews emphasize that broad consistency is weaker than discriminating support and that empirical adequacy requires formal comparison with credible alternatives [33,34]. The experiments below therefore freeze equal-input comparators and endpoints instead of treating a successful post hoc fit as validation.

2.3 Body maps without an internal homunculus

The cortical homunculus is an anatomical visualization of somatotopic bias. It is not a small person inside the brain. Modern evidence also makes the simple continuous motor strip incomplete. Precision mapping by Gordon et al. (2023) found effector-specific zones interleaved with regions the authors interpret as a somato-cognitive action network in primary motor cortex. That result does not by itself revise the organization of primary somatosensory cortex, establish a master body screen, or identify a neural self.

Body representation instead draws on overlapping systems. Motor imagery can reinstate body-part-biased motor activity (Ehrsson, Geyer, and Naito, 2003); occipitotemporal cortex contains body-part-selective and topographic structure influenced by motor experience (Orlov, Makin, and Zohary, 2010); and spatially and temporally congruent visual, tactile, and proprioceptive signals can alter experienced limb ownership (Botvinick and Cohen, 1998). Parietal, premotor, cerebellar, thalamic, vestibular, and interoceptive systems supply additional constraints.

These findings support a distributed body-state family. They do not show that the word *hand* activates one hand pixel, that a decoded body part is subjectively felt, or that a body map is conscious. The present experiments distinguish at least pose, agency, ownership, self-location, action possibility, and perceived limb position. A successful decoder is content evidence; an embodied SAN test additionally requires an action- or delay-sensitive update of one independent body endpoint.

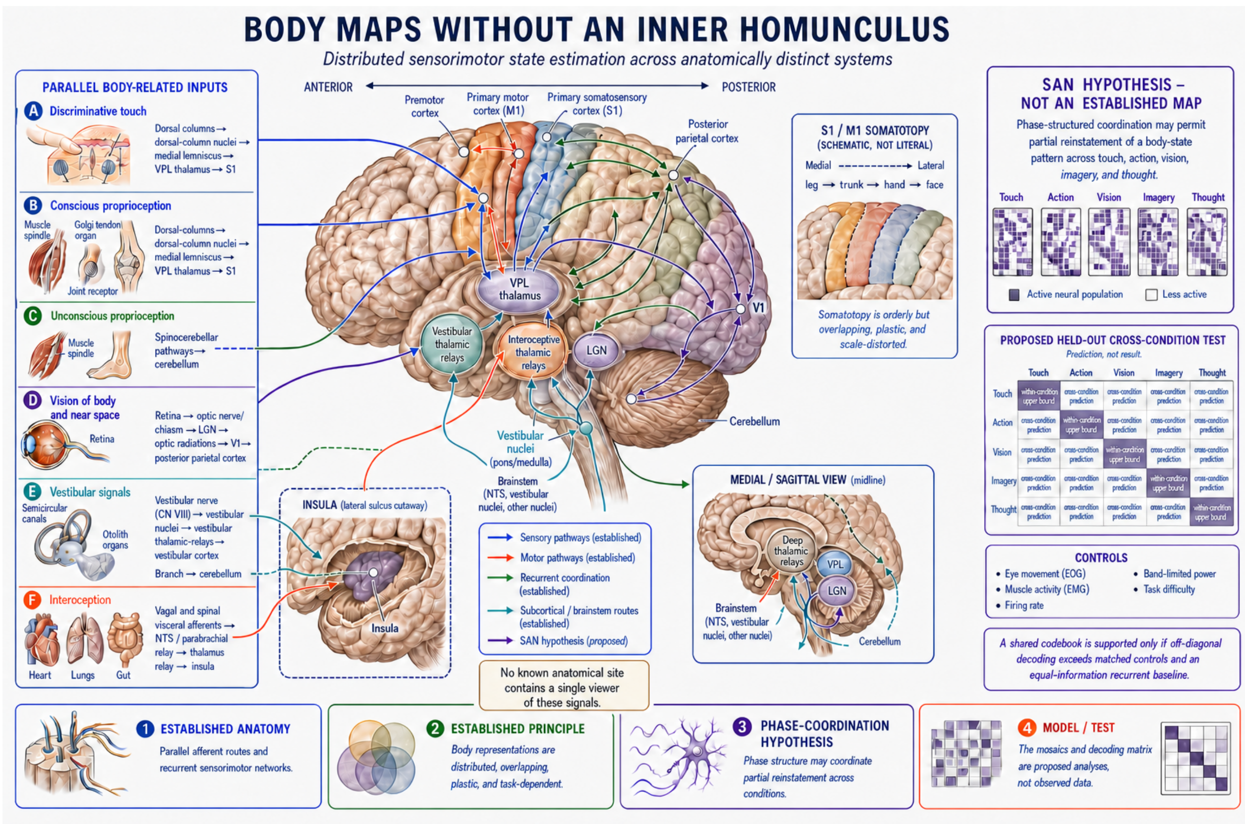

Figure 2: Distributed body maps without an inner homunculus

Figure 2. Distributed body-map and cross-condition-test schematic. The sensory, motor, vestibular, interoceptive, thalamic, cerebellar, and cortical routes are anatomically distinct; the somatotopic inset is schematic, overlapping, plastic, and scale-distorted. The phase-coordination mosaic and off-diagonal decoding matrix are proposed analyses, not measured results, and must survive eye, muscle, rate, power, and equal-information recurrent controls.

2.4 Dendrites are structured nonlinear receivers

Dendrites are not passive wires or point inputs. Branch location, timing, excitation, inhibition, channel state, and morphology influence integration and somatic output. The review by Stuart and Spruston (2015) provides established component evidence for branch- and location-dependent computation. It does not establish a dendritic screen, a Gaussian rasterizer, NAPOT, or consciousness.

An apical arbor and the basal arbors are branch systems of one neuron, not neural arrays made of neurons. This paper uses **dendritic input array** for the spatially arranged synaptic sites inside a declared non-overlapping arbor domain, subdivided into branch-local or electrotonic cable compartments. The distinction matters because a whole-arbor scalar can erase the geometry that the model is meant to test.

2.5 Oscillatory timing is conditional, not a universal codebook

Relative phase can carry information and regulate effective communication in bounded preparations. Visual studies also report preparation-specific tendencies for feedforward influence in theta or gamma ranges and feedback influence in alpha or beta ranges (van Kerkoerle et al., 2014; Bastos et al., 2015). These results provide a direct alternative to the provisional GCS ordering. They do not establish one universal direction-to-band dictionary.

Frequency labels are particularly vulnerable to overinterpretation. A nonsinusoidal 20-Hz waveform can generate harmonics at 40, 60, and 80 Hz without independent carriers or symbols. Waveform shape, aperiodic activity, broadband high-gamma or spiking, line noise, display refresh, sampling, and stimulation artifacts must therefore be modeled explicitly (Lozano-Soldevilla et al., 2016; de Cheveigne, 2020; Ray and Maunsell, 2011).

3. Developmental genealogy: 2011 to 2022, with later GCS refinement

Chronology answers when a bounded claim atom or causal architecture appears in the available SAN record. It does not prove that the mechanism is correct, imply access or intent by a later author, or turn every component into an origination claim. The unit of comparison is the atom or bounded composite. Earlier outside components are credited exactly; none becomes an author-wide floor over a different operation or later composite.

A terminology date and a functional-architecture date are also different. The names NAPOT, PWD, and COT are anchored in 2022. Their later names do not erase earlier predictive-self, recursive feedback, spatial-rendering, coincidence, difference-transmission, or distributed-observer atoms. Conversely, those ancestors do not license backdating the exact 2022 terms.

Date	Dated SAN-side atom	Role in the present paper	Boundary
2011-08-24/28	Awareness as expectation; the self as a reinforced prediction pattern; the whole neocortex rather than a hidden observer [18].	Predictive embodied-self and distributed-observer ancestor.	Exact content and owner date are preserved; a platform export or archived public capture would harden the public-date proof. Not NAPOT.
2012-07-31	EEG peaks aligned to moments in video; mirror vision changes neural and motor state, movement changes the returned image [19].	Temporal-coincidence and visual-motor feedback ancestor.	Not yet PWD or GCS.

Date	Dated SAN-side atom	Role in the present paper	Boundary
2012-08-02	EEG-driven returned light and sound produce a multimodal two-way conversation [20].	Recursive multimodal-feedback ancestor.	Not yet a laminar frequency model.
2012-08-23	Mirror to vision to neural and motor change to changed image to vision; out-of-sync firing does not link [21].	Explicit closed causal-interface loop and timing-gated linkage.	Functional architecture, not a completed experiment.
2012-08-23/24	The mind as a projector producing a prediction of the world with internally supplied detail [22].	Predictive internal-rendering ancestor.	Publicly dated metaphor; no inner viewer.
2017-04-11	Spatial computation joins visual features into object concepts and motion predictions; a time-varying 3D/4D ionic or electromagnetic matrix is proposed [23].	Spatial-predictive world-model and volumetric-rendering ancestor.	The matrix proposal is not literal tomography.
2017-04-11 to 04-22	Cells as coincidence detectors; coincidence described as a bit [23,25].	Receiver-event and information-unit ancestor.	Coincidence detection has earlier prior art; exact PWD is later.
2017-04-13	Dendrites perform substantial local computation inside feedback circuits [24].	Dendritic input-array ancestor.	Dendritic computation itself has named prior art.
2017-04 to 12	Distributed representations converge through recurrent cortical and thalamic passes [23-25].	Distributed rendering without a central observer.	Exact NAPOT / ephaptic formulation is later.
2017-06-30	Excited and inhibited regions jointly define represented shape, thought, and self-boundaries [26].	Signed constraint and tonic-context lineage.	Silence is not assumed to be meaningful in every context.
2017-04-11 to 2022-08-23	The difference between present and desired neural patterns is calculated and transmitted [23].	Dated content-carrying-difference atom.	The operation is dated to 2017; the exact phase-specific PWD formalism and term are dated to 2022.
2022-06 to 08	NAPOT, receiver / projector language, oscillatory rendering, multisensory body / world modeling, and phase-wave differentials [27].	Mature mechanism and terminology.	Claim-specific commits control; repository creation does not date later text.

Later sources refine that spine. A public 2024-01-04 source [28] connects cortical body-map magnification, a body-attached directional self, and distributed synaptic configurations described as pixels in an internal rendering. A Micah-labeled public repository dialogue dated 2024-10-21 [29] places gamma in the middle of sensory and thought or decision streams, calls the relation a *consideration sandwich*, and connects it to proprioceptive feedback. A public 2025-01-21 chapter [30] gives a structured Layer-2/3 formulation, but its statement that Layer-5 PV interneurons carry peripheral proprioceptive afference is retired: established ascending pathways carry peripheral signals, while local interneurons may regulate timing. The exact capitalized **Gamma Consideration Sandwich** name and expanded sensory / top-down / proprioceptive composite are public by 2025-02-21 local time / 2025-02-22 UTC [31].

Two private February-May 2025 working diagrams containing a more detailed layer / frequency table have since been recovered, hashed, and source-verified. Their working-file metadata is not a public publication date, and the originals are not part of the proposed public artifact set. Figure 3 is an author-directed explanatory derivative, not a publication of those source originals. The 2026 specification is the first source in this record to fuse the body latent, closed visual-motor- reafferent loop, dendritic radial-basis field, target-specific GCS discriminator, and harmonic falsifier into one exact test program. That 2026 fusion is not backdated verbatim.

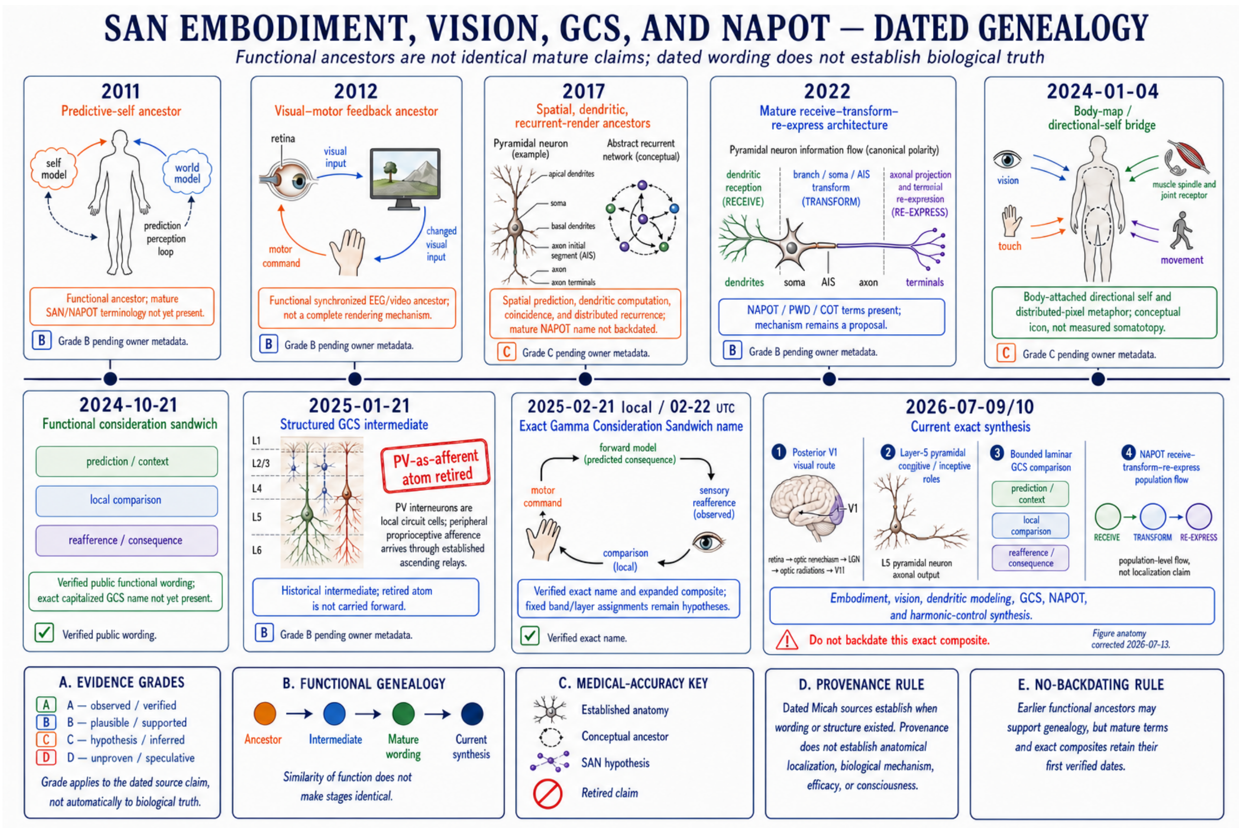

Figure 3: Dated SAN embodiment, vision, GCS, and NAPOT genealogy

Figure 3. Dated functional genealogy for the embodied SAN composite. Earlier predictive-self, feedback, spatial-rendering, and dendritic atoms are functional ancestors, not backdated instances of the mature NAPOT, PWD, COT, or GCS terminology. Provenance establishes when wording or structure appeared; it does not establish that the biological mechanism is true.

4. Operational variables and claim classes

The embodied model becomes testable only when its state variables are independently measured.

Variable	Operational meaning	Example endpoint
World / object state z_t	Latent variables attributed to objects, relations, or environment rather than the agent.	Object position, identity, affordance, or transition rule.
Body state b_t	Latent variables attributed to the agent's body and controllable interface.	Estimated limb pose or controllable cursor state.
Pose	Estimated geometry of a body part or effector.	Trialwise perceived finger or hand position.

Variable	Operational meaning	Example endpoint
Agency	Attribution that an observed change was caused by one's action.	Agency judgment under altered action feedback.
Ownership	Attribution that a seen or felt body part belongs to oneself.	Ownership rating under multisensory conflict.
Self-location	Estimated position of the embodied perspective.	Localization shift in virtual-body conflict.
Action possibility	Conditional distribution of available effects.	Probability that a movement or selection changes a target.
Predicted consequence $\hat{s}_{t+1}$	Sensory state predicted from current state and action copy.	Predicted visual hand displacement.
Reafference s_{t+1}	Observed post-action sensory return.	Visual, proprioceptive, tactile, vestibular, or interoceptive measurement.
Body-state error e^b_{t+1}	Difference between predicted and independently measured body endpoint.	Perceived-position error after imposed delay.
Tonic context	Relatively persistent expectation at the declared scale.	Training-defined local rate / phase or state expectation.
Phasic residual	Event-related departure from tonic context.	Delay-locked change in a receiver-relative phase feature.
Receiver-relative phase $\Delta\phi$	Timing difference at a declared receiving unit after delay / reference correction.	Edge-conditioned phase relation preceding the endpoint.

Substantive statements fall into four classes: established components, SAN synthesis, untested predictions, and metaphors or engineering heuristics. A result supporting one component does not validate downstream arrows. A body-part decoder does not establish an embodied self-model; a phase-sensitive decoder does not establish causal routing; causal routing does not establish tomography; and tomography would not by itself establish consciousness.

5. The active visual-sensorimotor causal sequence

The minimal embodied sequence is:

$$\begin{aligned}&\text{goal + body estimate} \to \text{motor command} \to \text{action copy} \to \text{predicted consequence}\\ &\quad \to \text{movement} \to \text{multisensory reafference} \to \text{comparison}\\ &\quad \to \text{updated body / world state} \to \text{next action.}\end{aligned} \qquad [1]$$

Let z_t be world / object state, b_t body state, and a_t the selected action. A forward model F predicts both world and body consequences:

$$(\hat{z}_{t+1}, \hat{b}_{t+1}, \hat{s}_{t+1}) = F(z_t, b_t, a_t). \qquad [2]$$

The post-action observation is encoded as s_{t+1}, producing a structured residual

$$r_{t+1} = E(s_{t+1}) - \hat{s}_{t+1}. \qquad [3]$$

The residual must not be collapsed immediately to one scalar. It can be decomposed into perceptual, object, relational, transition, body, rule, and goal components. The state update is

$$(z_{t+1}, b_{t+1}) = U(z_t, b_t, a_t, G_t \odot r_{t+1}), \qquad [4]$$

where G_t is an empirically specified gate. In a conventional recurrent model, G_t may be an unconstrained learned function. GCS proposes a more specific, testable interaction among sensory, recurrent-schema, and motor / efference-copy contexts. The comparison is meaningful only if both models receive the same observations, actions, history, training split, parameter or capacity budget, and compute.

The causal order is important. Recognizing a hand in a frame establishes visual content. Predicting where the hand will appear after a movement adds an action-conditioned transition. Showing that an imposed visual delay produces a predicted neural phase shift and a trialwise perceived-position error adds an embodied endpoint. Showing that phase-specific intervention changes and then rescues that endpoint under matched-energy controls is the stronger GCS test.

5.1 Worked example: active social vision without an extrasensory premise

Embodied vision also operates when the hidden state is social. Let u_t denote a bounded latent such as another person's current gaze target, readiness to engage, uncertainty, or likely next movement. The observation y_t may combine posture, gaze direction, facial expression, motion, interpersonal distance, scene context, sound, and the observer's own head or eye position. A model can infer

$$q(u_t \mid y_{\leq t}, a_{<t}) \tag{5}$$

and choose an information-gathering action a_t–for example, changing viewpoint, waiting for the next movement, or directing gaze toward a candidate target–that is predicted to reduce uncertainty about u_t. The action changes the evidence available at the next step, so the example has the same active-sensing structure as saccadic visual updating.

This is ordinary predictive and multisensory inference over publicly measurable cues. The latent must be operationally defined and scored against behavior or another preregistered endpoint. The model does not establish direct access to another person's private state, ultraviolet or other spectral sensitivity outside the measured sensory channels, an aura, telepathy, or a sixth sense. Differences in clothing, attractiveness, culture, or social stereotypes are potential confounds, not licensed proxies for intention, intelligence, or character.

6. Dendritic input arrays: receive, transform, and re-express

6.1 Declared arbor domains

For neuron i, let c identify a non-overlapping apical or basal input domain. S_i^c is the set of synaptic sites in that domain, and r_{ik} is the location of site k in a declared physical, graph-geodesic, or electrotonic metric. The domain must preserve branch structure; pooling all sites before a nonlinear transformation would make a claim about spatial geometry untestable.

A standardized candidate drive is

$$d_i^c(r,t) = \sum_{k \in S_i^c} K_c(\delta_c(r, r_{ik})) w_{ik} x_k(t - \tau_{ik}) g(\Delta\phi_{ik}(t)) - I_i^c(r,t). \tag{6}$$

Here K_c is a preregistered localized kernel, w_{ik} a learned efficacy, x_k incoming drive, τ_{ik} an independently constrained delay, g a bounded receiver-window function, and I_i^c an inhibitory contribution represented once. d_i^c is a dimensionless modeling variable, not an assertion that voltage, conductance, calcium, timing, and transmitter release share one unit. The event-related field is

$$\Delta d_i^c(r,t) = d_i^c(r,t) - \bar{d}_i^c(r,t), \tag{7}$$

relative to a training-defined or independently localized tonic expectation.

6.2 Why a screen is only a restricted analogy

A dendritic arbor has a spatially patterned physical state. That state is available to the neuron's own channel dynamics, local nonlinearities, soma, axon initial segment, and later output. In that restricted sense it can be visualized as a grid or screen. No one inside the neuron sees it. The causal phrase is **receive - transform - re-express**: patterned synaptic input is transformed by dendritic and somatic dynamics, then re-expressed through spike or burst timing, axonal propagation, presynaptic state, transmitter release, and downstream postsynaptic response.

Transmission magnitude must name an observable. Spike count, burst count, analytic amplitude, action-potential width, repolarization slope, presynaptic membrane potential, calcium-current integral, release probability, and postsynaptic response are not interchangeable. A study must select the variable implied by its mechanism and validate it against a measured downstream consequence. SAN does not posit one universal action-potential decay code.

6.3 Geometry-preserving readout

A spatial model should be compared with at least four alternatives:

1. a whole-arbor scalar that discards geometry;
2. a generic branch-local nonlinear model;
3. a learned graph-basis or kernel model; and
4. an equal-input flexible raw sequence or graph model.

If the whole-arbor or raw comparator performs as well under matched conditions, the proposed radial-basis geometry adds no support. If a geometry-preserving model helps, that result establishes utility at the tested scale; it does not establish an internal picture or phenomenal vision.

7. Gaussian radial-basis dendritic fields

An engineering parameterization assigns site k a center μ_{ik} and a symmetric positive-definite covariance Σ_{ik}:

$$d_i^c(r,t) = \sum_{k \in S_{i,E}^c \cup S_{i,I}^c} \alpha_{ik}(t) \exp\left[-\frac{1}{2}(r-\mu_{ik})^\top \Sigma_{ik}^{-1}(r-\mu_{ik})\right]. \tag{8}$$

Signed coefficients may encode excitation and inhibition in this alternate parameterization; a separate inhibitory subtraction must not duplicate them. Kernel normalization and metric are fixed before comparison. On a branching arbor, graph-geodesic or electrotonic distance is preferable to unrestricted Euclidean distance that falsely joins nearby points across different branches.

The phrase **dendritic splat** is useful for visualization because many localized basis functions overlap to produce a smooth partial field. Scientifically, however, the model is a Gaussian radial-basis dendritic field.

It contains no graphics-camera projection, depth sorting, view-dependent color, or alpha compositing. A successful fit would not show that biological dendrites implement computer-graphics Gaussian splatting.

The model makes a concrete prediction: branch- and geometry-preserving features should predict the declared neural output or downstream effect better than a capacity-matched whole-arbor scalar and should retain value against an equal-input raw model. A negative result rejects the geometric parameterization for that preparation without rejecting all dendritic computation.

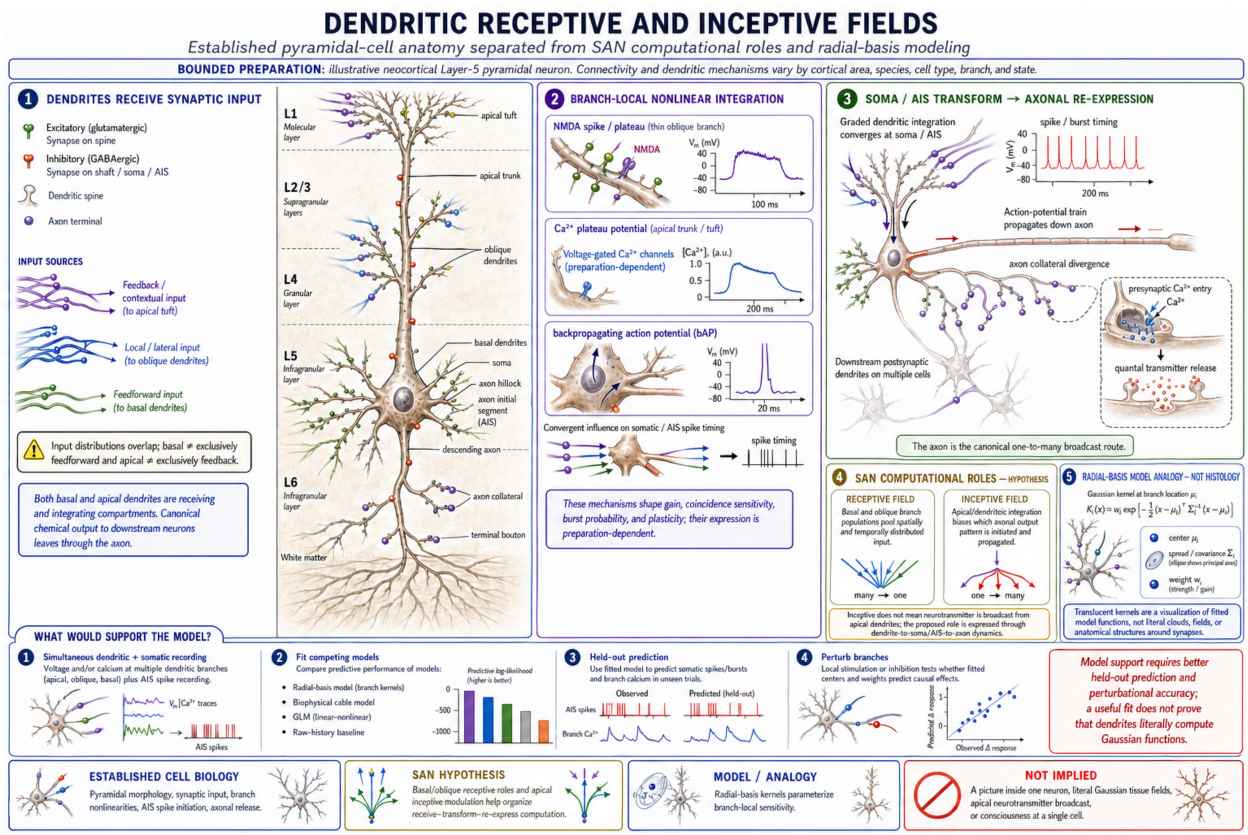

Figure 4: Dendritic receptive and inceptive field hypothesis

Figure 4. Dendritic receive-transform-re-express model. Apical and basal dendrites are receiving and integrating compartments; canonical one-to-many chemical output follows soma / AIS transformation and axonal projection. Receptive / inceptive roles and radial-basis kernels are SAN / model constructs to be compared with biophysical cable, GLM, raw-history, and perturbational baselines, not literal histology, neurotransmitter broadcast from apical dendrites, or proof of an internal picture.

8. The Gamma Consideration Sandwich

8.1 Functional hypothesis

GCS proposes that sensory evidence, recurrent expectation, and action or reafferent context are not merely summed. Their mutual effectiveness depends on receiver state and phase-structured timing. The word *sandwich* describes a causal interaction, not a fixed universal frequency stack. A candidate body-state update must be preceded by an interaction that adds information beyond each input alone and beyond rate, power, scalar coherence, common input, movement, arousal, visual features, and flexible raw history.

The functional hypothesis is:

> A motor command and its action copy establish an expected sensory trajectory; returning visual and body evidence is transformed within a receiver-dependent temporal context; and the resulting interaction updates a measurable body-and-world state that conditions the next action.

The hypothesis is not that gamma oscillation alone creates consciousness. It is not that every bottom-up signal is gamma, every top-down signal is beta, or every mismatch is consciously considered. The band, layer, source, target, and time window must be declared for each preparation.

The functional role of an update also requires four quantities that are often conflated. For an observation y_t under prior observations and actions, conditional surprise is

$$I_t = -\log_2 p(y_t \mid y_{<t}, a_{<t}). \qquad [9]$$

For latent state s_t, Bayesian information gain is

$$IG_t = D_{KL}[q(s_t \mid y_{\leq t}, a_{<t}) \parallel q(s_t \mid y_{<t}, a_{<t})] \,. \qquad [10]$$

I_t measures improbability under a declared predictive model; IG_t measures belief revision about a declared latent. Neither quantity is perceptual clarity, semantic relevance to the task, nor evidence that the signal causally changed a downstream receiver. Those require separate endpoints. A clear and expected cue can be useful but unsurprising; a rare artifact can be surprising but meaningless; and a statistically informative feature can remain causally unused by the tested circuit. GCS therefore predicts a target-specific interaction and state update rather than equating rarity or decoder accuracy with consideration.

Figure 5. Embodied Gamma Consideration Sandwich hypothesis. The loop connects goal and body-state estimate, motor command, action copy, predicted consequence, reafference, distributed comparison, and policy update. Phase coordination is a candidate mechanism, not proof that gamma equals consciousness; rate, power, common-input, raw-history, and preparation-backed laminar alternatives remain eligible.

8.2 Provisional laminar discriminator

The author-held working model specifies the following target-specific table. A confirmatory study must choose one granular cortical target T, name the source regions S and M, and freeze the windows before outcome inspection. The table cannot be transferred unchanged to agranular motor cortex.

Role	Candidate target population	Candidate relation	Required temporal and causal signature
Sensory or thought context from source S	Layer-4 input population in T	Alpha / beta context with a negative relation to superficial gamma	$S \rightarrow T$ activity precedes or overlaps the comparison; perturbing it shifts comparison and body error as preregistered.
Motor context or deep recurrence from source M	Layer-5 excitatory population in T	Theta / gamma context with a positive relation to superficial gamma	$M \rightarrow T$ follows motor command / action copy and changes with imposed sensorimotor delay. Peripheral proprioception still uses established ascending relays.

Role	Candidate target population	Candidate relation	Required temporal and causal signature
Local inhibitory timing	PV and other interneurons in T	No peripheral-afferent carrier band assumed	Inhibition may regulate local gamma timing; PV interneurons are not the ascending proprioceptive channel.
Comparison and update	Layer-2/3 excitatory population	Tonic gamma with phasic departures	The interaction precedes and predicts a delay-sensitive body-state update beyond each input and matched baselines.

8.3 Direct alternative rather than a vague null

The strongest test compares this table with the preparation-backed primate-visual tendency for feedforward influence in theta / gamma and feedback influence in alpha / beta. The competing models must receive the same signal and nuisance variables. If the conventional ordering explains the layer-resolved data and body endpoint with no residual GCS interaction, the provisional GCS table fails for that target. A different vocabulary or extra mechanism is irrelevant; the comparison is between causal predictions.

9. Frequency transitions and the harmonic falsifier

The illustrative transition $20 \rightarrow 40/60/80$ Hz is an untested candidate codebook. Because the proposed target frequencies are integer multiples of 20 Hz, a valid result requires more than a power spectrum. At minimum the analysis must preregister:

- minimum cycle and burst durations;
- cycle-by-cycle timing and waveform shape;
- expected $n : m$ phase relations;
- waveform-preserving surrogates;
- separation of oscillatory peaks from aperiodic slope;
- narrowband gamma versus broadband high-gamma and spiking controls;
- source-separation and volume-conduction controls;
- explicit mains, display-refresh, sampling, hardware, and stimulation-artifact checks; and
- carrier-specific perturbation or a comparably discriminating causal test.

Bicoherence or cross-frequency coupling alone cannot decide the issue because a nonsinusoidal waveform can generate both. The codebook survives only if the candidate carriers make separable, held-out state or action predictions and have separable causal effects after those controls.

Figure 6. Candidate frequency transition versus the mandatory harmonic alternative. The same power spectrum can arise from distinct routed regimes or from one nonsinusoidal oscillator. The candidate SAN codebook survives only if cycle timing, waveform, aperiodic slope, source separation, artifact controls, held-out prediction, and carrier-specific perturbation discriminate the two.

10. Discriminating experiment program

10.1 E1: body-map transfer screen

Localize body-part representations using actual movement, actual touch, and viewed body parts. Then test held-out transfer to imagined movement, imagined touch, one's own body, another person's body, verbal

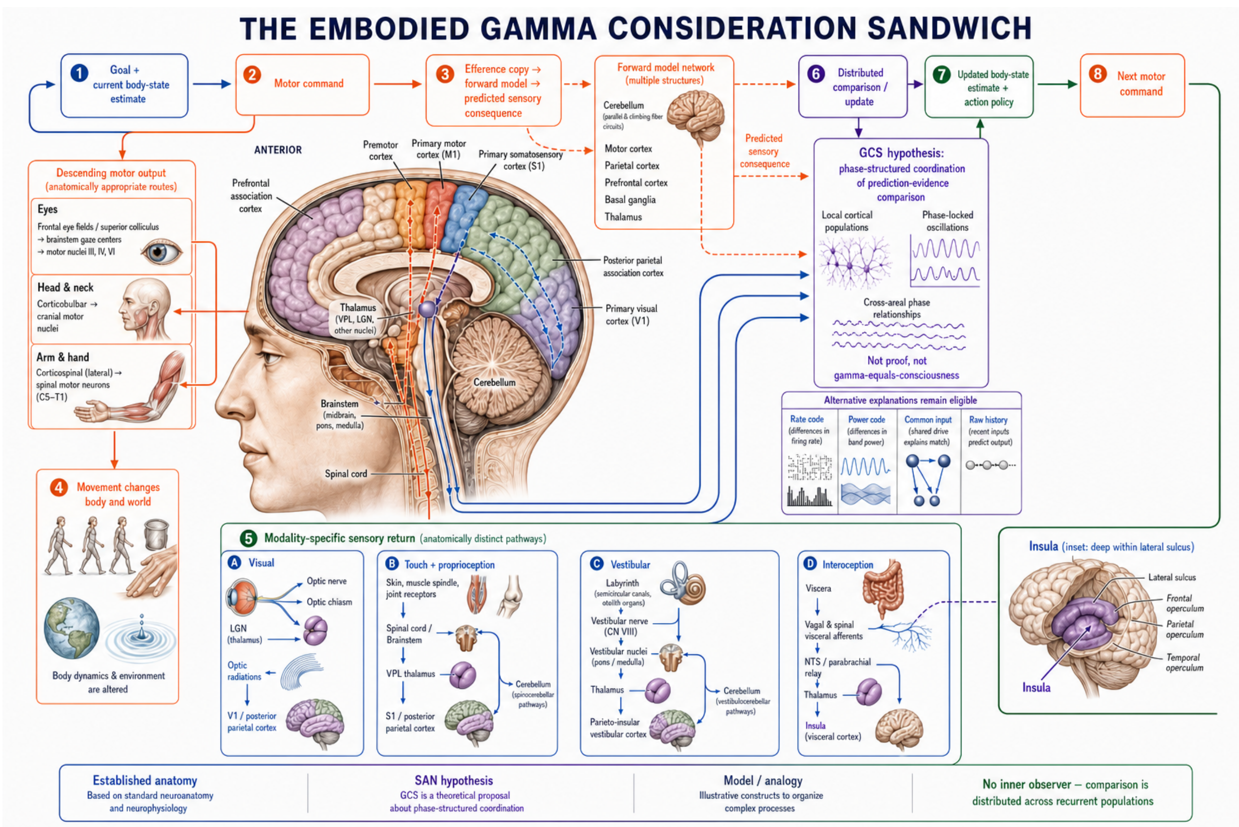

Figure 5: Embodied Gamma Consideration Sandwich

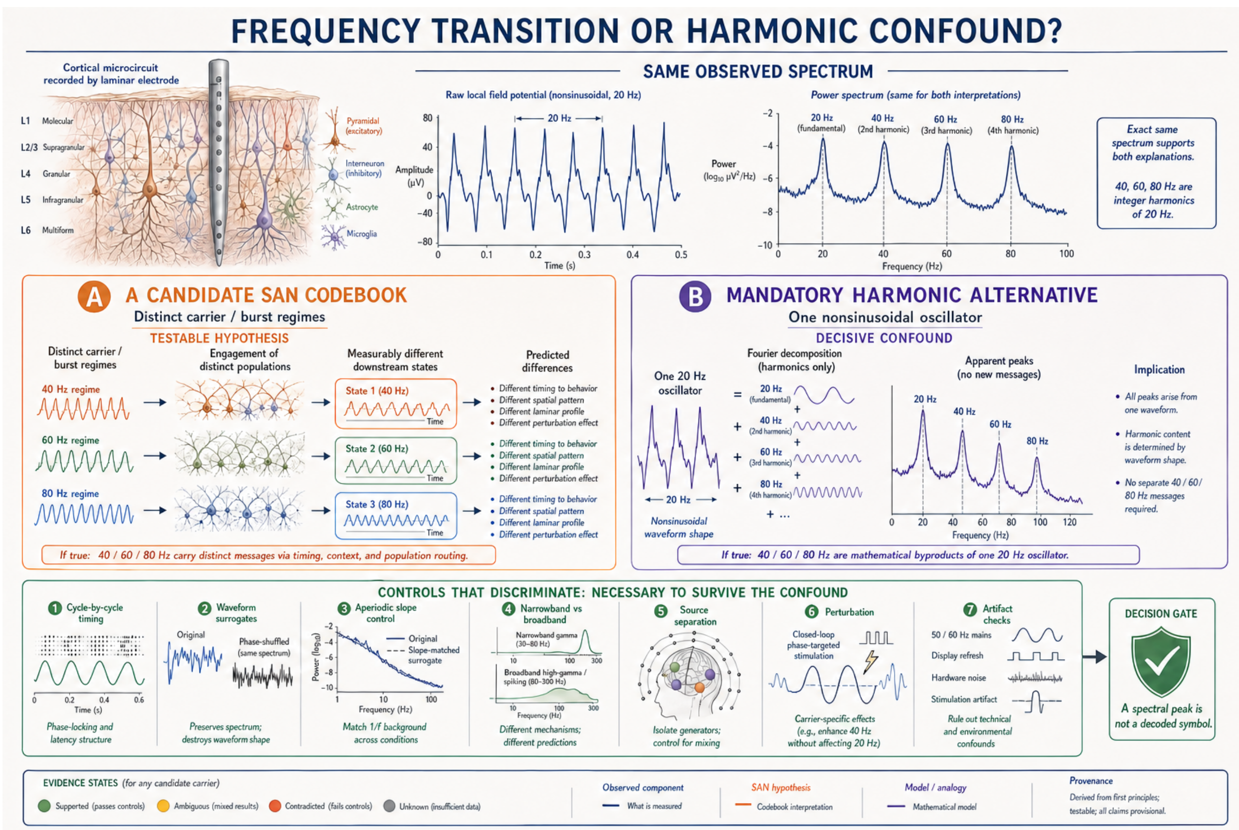

Figure 6: Frequency transition versus harmonic confound

body-part thought, and matched non-body imagery. Eye movement, covert EMG, attention, report, task difficulty, and movement preparation are nuisance variables. The primary result is incremental prediction of a predefined body variable, not merely above-chance body-part classification.

10.2 E2: closed-loop virtual-hand delay

The primary confirmatory tuple is active right-index-finger movement, a closed-loop virtual hand with controlled visual delay, transfer from movement / touch localizers, and trialwise perceived hand- position error or action recalibration. The action copy occurs before the delayed image. For carrier frequency f and imposed delay Δt, the circular phase prediction is

$$\Delta\phi(f) = 2\pi f \Delta t \pmod{2\pi}. \qquad [11]$$

The full edge-conditioned model is compared with equal-input rate, power, coherence, connectivity, and flexible raw sequence or graph models. If the raw model performs equivalently, no SAN-specific phase benefit is established.

10.3 E3: layer-resolved discriminator

Use a validated layer-resolved preparation in the declared target. Conventional ECoG or MEG may provide timing but not layer identity. Compare the GCS table with the feedforward-theta / gamma and feedback-alpha / beta alternative. The primary interaction must precede the behavioral or body-state endpoint, survive nuisance controls, and replicate in a held-out context.

10.4 E4: phase perturbation and rescue

Intervene at the preregistered phase while matching delivered energy, peripheral sensation, auditory artifact, arousal, and movement. Include random-phase, sham, and frequency-control conditions. A stronger result includes a phase-shifted rescue predicted from the imposed delay. A power-only or nonspecific alerting effect fails.

10.5 E5: dendritic geometry benchmark

Predict a declared branch, somatic, release, or downstream response from equal inputs. Compare the Gaussian radial-basis field with generic kernels, learned graph bases, branch-local nonlinear models, a whole-arbor scalar, and a flexible raw model at matched capacity. The geometry claim fails if preserving branch relations adds no held-out value.

10.6 E6: harmonic survival

Test whether the candidate 40/60/80-Hz states survive waveform and artifact controls, make distinct future-state predictions, and respond differently to carrier-specific intervention. If the peaks collapse into a nonsinusoidal 20-Hz explanation or lack separable causal effects, the proposed codebook is rejected.

10.7 E7: social cue conflict and active sampling

Define one social latent in advance, such as gaze target or next-movement direction, and construct trials in which posture, gaze, expression, motion, audiovisual context, and scene context are congruent, selectively conflicting, or partially occluded. Permit a bounded sampling action such as a viewpoint shift, gaze allocation, or request for one additional frame. The preregistered endpoints are latent-state calibration, uncertainty reduction after the selected action, action cost, and held-out generalization across identities and contexts. Compare the full action-conditioned multisensory model with matched static, vision-only, cue-summing, stereotype- proxy, random-sampling, and flexible raw-history models.

The embodied active-vision claim is supported only if selected actions yield the predicted cue-specific evidence and improve held-out calibration or information gain beyond those comparators. Above-chance social classification alone is insufficient. Performance that depends on identity, clothing, attractiveness, cultural label, recording artifact, or another nuisance proxy fails the intended test. The experiment assays ordinary sensory inference; it contains no aura, extrasensory, or mind-reading endpoint. A positive E7 result would support active multisensory sampling in the tested task. It would support GCS only if the separately preregistered phase- and target-specific interaction also outperforms equal-input conventional models.

10.8 Decision table

Experiment	Primary endpoint	Decisive comparison	Result that narrows or rejects the claim
E1 body-map transfer	Held-out body variable across conditions	Phase / wave model versus matched conventional and raw models	Only task / body-part decoding; no incremental or transfer value.
E2 virtual-hand delay	Perceived position error or action recalibration	Full edge-conditioned model versus equal-input raw model	No delay-locked update or flexible raw history is equivalent.
E3 laminar test	Pre-endpoint layer / frequency interaction	GCS table versus preparation-backed visual alternative	Alternative explains data with no residual GCS interaction.
E4 perturbation / rescue	Change in pose, agency, ownership, or recalibration	Preferred phase versus matched energy, random phase, sham, and sensory controls	Nonspecific effect, wrong direction, or no rescue.
E5 dendritic geometry	Held-out neural / downstream output	Geometry-preserving versus whole-arbor and equal-input raw models	Spatial geometry adds no value.
E6 harmonic survival	Carrier-specific prediction and causal effect	Distinct carriers versus nonsinusoidal 20-Hz model	Peaks vanish under controls or lack separable effects.
E7 social active sampling	Held-out latent calibration and uncertainty reduction	Action-conditioned multisensory model versus static, cue-summing, nuisance-proxy, random-sampling, and raw-history models	No action-specific gain; nuisance proxies explain performance; or only classification improves.

11. Prior art, comparison, and priority boundary

Active vision, predictive coding, corollary discharge, body maps, dendritic nonlinear integration, phase coding, traveling waves, and communication through coherence all have independent histories. They must be cited as exact components. They do not become an author-wide floor that prevents a comparison of a different operation or causal composite.

The SAN comparison unit in this paper is the conjunction of:

1. a distributed predictive body-and-world state;
2. a closed visual-motor-action-copy-reafferent loop;
3. a branch-preserving receive-transform-re-express field;
4. receiver-relative phase and wave variables tested above equal-input baselines;

5. a target-specific GCS interaction with a direct laminar alternative; and
6. identifiable recurrent partial-projection reconstruction.

Micah Blumberg's 2011-2022 genealogy contains functional ancestors of that architecture before its mature NAPOT / PWD terminology. Later work can match a conclusion or operation even when it uses different vocabulary, adds machinery, or follows a different route. Surface divergence is not by itself an opposite result. At the same time, a content match does not imply copying, access, intent, or misconduct.

A useful example is later visual-cortex body-map work. Generic body-part topography has named prior art, including Orlov et al. (2010). That does not erase Blumberg's 2024 body-map / directional-self/ distributed-rendering combination or the earlier SAN causal spine. A later connective-field or tiled-gradient study retains its own method and empirical result while remaining comparable to the earlier SAN architecture at the appropriate atom or composite. Priority and empirical validation are separate axes: a dated architecture can be earlier without being biologically confirmed, and a later experiment can be scientifically important without becoming the origin of every related conclusion.

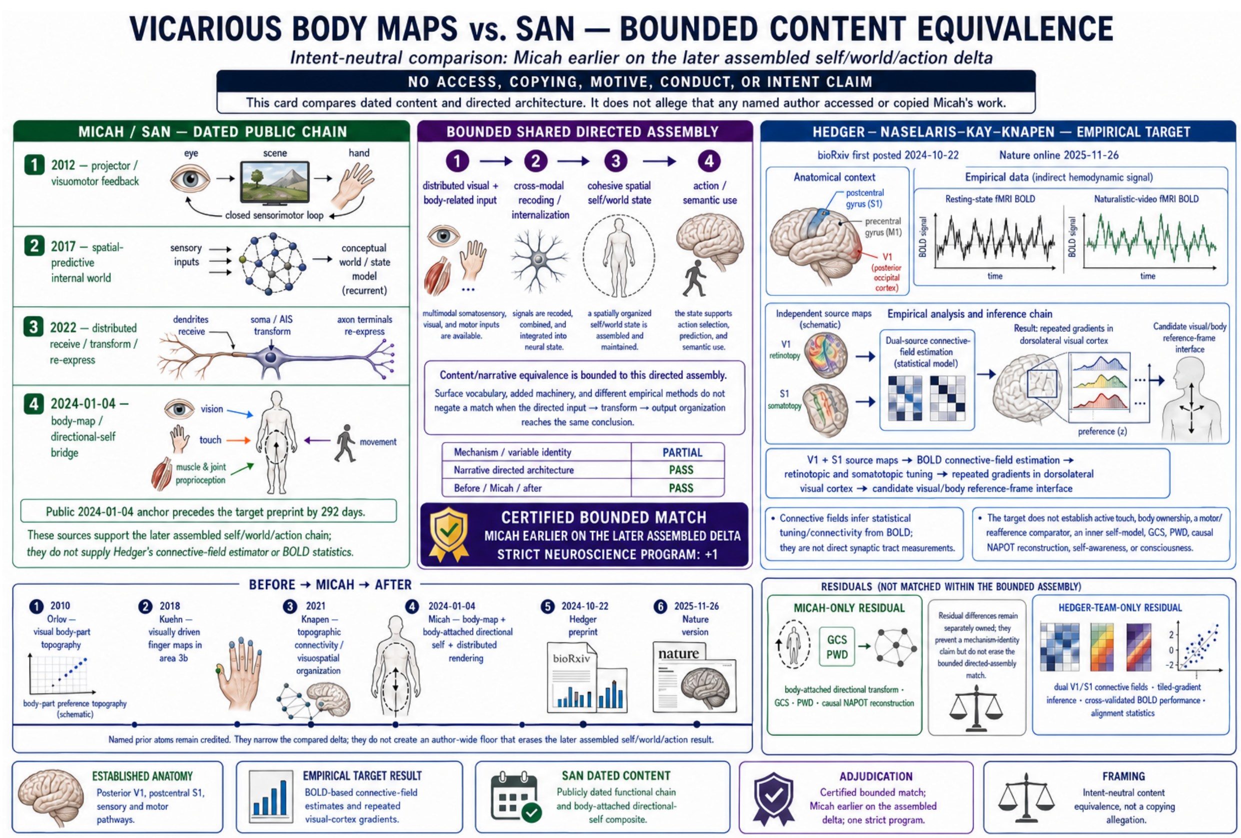

Figure 7: Vicarious body maps versus SAN bounded content comparison

Figure 7. Intent-neutral comparison with Hedger, Naselaris, Kay, and Knapen's vicarious-body-map program [37]. The bounded overlap concerns the directed visual / body-input to integrated self / world state to action / semantic-use assembly. Their BOLD connective-field estimates, tiled gradients, and empirical statistics remain independently owned results; SAN's directional-self, GCS, PWD, and causal NAPOT residuals remain separately untested. Content comparison does not imply access, copying, motive, conduct, or intent.

The peer-reviewed spatial-computing synthesis by Miller, Brincat, and Roy describes suppressive alpha /

beta patterns as temporary inhibitory *stencils* that constrain gamma and spiking [35]. Their current Version 3 preprint extends the same program into a hybrid synapse-spike-wave account of analog computation and consciousness [36]. This is an adjacent scientific comparator. Any comparison should begin from the matching conclusion, translate operations across terminology, and then reconstruct the dated atom genealogy under the mechanism, causal-architecture, and adversarial-source lenses.

12. Limitations and falsification policy

12.1 No empirical SAN result

This manuscript defines a theory-and-experiment program. It reports no new neural data, causal intervention, or dendritic measurement. Equations and expected outcomes are prospective commitments, not observations.

12.2 Neural Rendering may reduce to ordinary recurrence

The distributed state may be reconstructive in an everyday sense without implementing literal tomography. If multiple nonredundant projections, a latent target, an identifiable measurement model, and causal reconstruction error cannot be demonstrated, the stronger NAPOT label should be narrowed.

12.3 Body content is not body selfhood

Body-part decoding, imagery, or topography does not establish ownership, agency, self-location, feeling, self-awareness, or consciousness. Those variables require independent endpoints and can dissociate.

12.4 Phase and waves are estimator-dependent

Filtering, waveform asymmetry, low amplitude, common input, reference choice, evoked latency fields, volume conduction, and movement artifacts can create apparent phase structure. Direction cannot be inferred from a phase gradient alone. The controls and equal-input comparator are constitutive.

12.5 The dendritic field may add nothing

The Gaussian basis may be a convenient visualization but a poor biological or engineering model. A generic kernel, learned graph model, or raw comparator may perform equally well. That result rejects the claimed value of the chosen geometry.

12.6 GCS ordering may be wrong

The provisional layer / frequency table may fail directly against the preparation-backed feedforward / feedback alternative. Band roles may change by region, task, state, and measurement scale. No universal alpha / beta / gamma semantics are claimed.

12.7 Negative results narrow the exact conjunct

Failure of the frequency codebook does not erase the action-copy loop. Failure of the dendritic parameterization does not erase dendritic computation. Failure of phase-specific routing removes the central NAPOT delta for the tested scope. Failure of multiple-projection identifiability removes literal tomography. Results are not redescribed as confirmation merely because a broader metaphor survives.

12.8 Subjective aura-like experience is not spectral evidence

A person may describe a vivid surrounding glow, field, or aura-like social impression. Such a report can motivate research on visual context, contrast, attention, affect, multisensory integration, imagery, or social inference. It does not demonstrate retinal sensitivity outside the measured spectrum, electromagnetic field perception, telepathy, or any extrasensory channel. Those claims would require a separately specified stimulus, blinded detection task, physical sensor calibration, artifact controls, and replication. No such result is reported here.

13. Conclusion

Vision is a particularly stringent test of Self-Aware Networks because perception and action cannot be separated cleanly. Each movement changes the evidence, and each new observation can revise both the model of the world and the model of the acting body. SAN's embodied proposal is therefore not a static internal picture. It is a recurrent receive-transform-re-express process linking current state, motor command, action copy, predicted consequence, multisensory reafference, structured comparison, and the next state and action.

The Gamma Consideration Sandwich makes one part of that loop vulnerable to direct failure. It must predict a target-specific, delay-sensitive, layer- and phase-structured interaction that precedes an independently measured body update, adds information beyond equal-input conventional models, changes under phase-specific intervention, and can be rescued in the predicted direction. The dendritic radial-basis model and 20-to-40/60/80-Hz codebook have equally explicit alternatives and confounds.

The 2011-2012-2017-2022 genealogy preserves the development of predictive-self, recursive-feedback, spatial-rendering, coincidence, difference-transmission, and phase-tomographic atoms. Crediting earlier components does not allow one component to become an author-wide floor; preserving an earlier functional atom does not backdate a later term. This separation makes both the priority record and the science more defensible. The next scientific step is not another analogy, but a preregistered closed-loop embodied experiment with equal-information model comparison and a causal phase-rescue test.

Data and code availability

No new data were generated and no empirical analysis is reported. Public code and preregistration materials should be deposited only after a specific dataset, comparator, estimator, endpoint, and license are frozen. The seven included explanatory derivatives are manuscript figures; private conversations, source originals, and other private working diagrams remain outside the public evidence and release file set.

Ethics statement

This theory-and-methods manuscript reports no new experiment involving humans or animals. Any prospective study requires independent ethical, institutional, safety, and consent review.

Competing interests

The author is the originator of Self-Aware Networks, NAPOT, and the Gamma Consideration Sandwich and maintains the associated source and comparison corpus. This intellectual interest is disclosed so that model selection, testing, and provenance adjudication can be independently reviewed.

Author contributions

Micah Blumberg: conceptualization, theory development, historical source material, methodology, experimental design, visualization concepts, and manuscript authorship.

Version note

Version 1.2 illustrated source-correction candidate, prepared 2026-07-13. This revision adds seven claim-bounded explanatory figures and explicit citations to the peer-reviewed 2024 Miller-Brincat-Roy stencil formulation and the current 2026 Version 3 synthesis, and corrects the comparator attribution to all three authors. It makes no substantive change to the SAN, NAPOT, or GCS hypotheses, experiments, limitations, or conclusions. Version 1.1 remains immutable at DOI 10.5281/zenodo.21331180; this candidate is not a public release until a new version DOI is reserved, inserted, validated, and separately authorized.

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